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Division: A2

ASSIGNMENT No: 6

TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives: To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory: An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling outcome .

Exercise

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Program:

```
File Edit Selection View Go Run Terminal Help
interface Action Untitled-1
1 interface Action
2 {
3     void perform();
4 }
5
6 class Vehicle
7 {
8     String name = "Honda City";
9     int speed = 120;
10
11     class Details
12     {
13         void display()
14         {
15             System.out.println("Vehicle Name : " + name);
16             System.out.println("Speed : " + speed + " km/h");
17         }
18     }
19 }
20
21 public class Main
22 {
23     public static void main(String[] args)
24     {
25         Vehicle v = new Vehicle();
26
27         Vehicle.Details d = v.new Details();
28         d.display();
29
30         Action a = new Action()
31         {
32             public void perform()
33             {
34                 System.out.println("Vehicle is moving.");
35             }
36         };
37
38         a.perform();
39     }
40 }
```

Output:

```
File Edit Selection View Go Run T
PROBLEMS 15 OUTPUT DEBUG CONSOLE TERM
PS C:\Users\Ayush\projects> & 'C:\User
eStorage\44eea058444828944518bac730833f
Vehicle Name : Honda City
Speed : 120 km/h
Vehicle is moving.
PS C:\Users\Ayush\projects>
```

2. Develop a Food Delivery Application where an inner class handles order details and anymous classes handle delivery status updates.

Program:

```
1  interface Delivery
2  {
3      void status();
4  }
5
6  class FoodDelivery
7  {
8      String order = "Veg Pizza";
9      int amount = 450;
10
11     class OrderDetails
12     {
13         void display()
14         {
15             System.out.println("Order : " + order);
16             System.out.println("Amount : Rs." + amount);
17         }
18     }
19 }
20
21 public class Main3
22 {
23     Run main | Debug main | Run | Debug
24     public static void main(String[] args)
25     {
26         FoodDelivery fd = new FoodDelivery();
27         FoodDelivery.OrderDetails od = fd.new OrderDetails();
28         od.display();
29
30         Delivery d = new Delivery()
31         {
32             public void status()
33             {
34                 System.out.println("Order Delivered Successfully.");
35             }
36         };
37
38         d.status();
39     }
40 }
```

Output:

```
File Edit Selection View Go
PROBLEMS 17 OUTPUT DEBUG CONSOLE
PS C:\Users\Ayush\projects> &
eStorage\44eea058444828944518ba
Order : Veg Pizza
Amount : Rs.450
Order Delivered Successfully.
PS C:\Users\Ayush\projects>
```